LUND UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Number Theory
Friday January 12, 2018
Duration: 8.00–13.00

Solutions may of course be presented in Swedish! The sheets of the department should be used. Write your initials and at most one solution on each sheet handed in. Give careful motivations to your solutions!

1. Find the last digit of 7^{139} and 13^{201}.

2. Find an integer having at least three different representations as a sum of two squares, disregarding signs and the order of the summands.

3. Let p be an odd prime and let $n \geq 2$. Show that for $a \in \mathbb{Z}$,

$$a^p \equiv 1 \pmod{p^n} \iff a \equiv 1 \pmod{p^{n-1}}.$$

4. Find all integers n such that

$$\phi(n) = \frac{n}{3}.$$

(Here ϕ denotes Euler's phi-function.)

5. Let $m \in \mathbb{Z}$, $3 \nmid m$. Prove that there is at least one prime p dividing $4m^2 + 3$ such that $p \equiv 7 \pmod{12}$.

6. Prove that for a prime $p \geq 5$,

$$\sum_{j=1}^{p-1} \frac{1}{j^2} \equiv 0 \pmod{p}$$

in the sense that p divides the numerator of the left hand side.